Experience-Centered Spatial Simulation: A Computational Framework for Modeling Stakeholder Experience with Generative Agents

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Architectural design requires consideration of experiential differences across stakeholders, but such feedback is difficult to gather from people or to model computationally before a space is built. We introduce *Experience-Centered Spatial Simulation* (ECSS), a computational framework that uses generative agents to model spatial experience as an evolving interaction between environmental information and individual profiles. ECSS decomposes experience formation into explicit stages of sensory perception, environmental evaluation, and personal appraisal, that inform interpretations and actions over time. We instantiate the framework in a 2D spatial simulator and evaluate it through five comparative scenarios varying layout, prior memory, stakeholder role, weather, and occupancy. Results show interpretable differences across conditions, demonstrating the framework's ability to differentiate simulated spatial experiences. We conclude by discussing the framework's challenges, scope, and limitations, positioning ECSS as an augmentative design tool that complements architectural design without replacing stakeholder participation or designer judgment.

CCS Concepts: • **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Additional Key Words and Phrases: Do, Not, Use, This, Code, Put, the, Correct, Terms, for, Your, Paper

1 Introduction

Designing effectively for future users based on incomplete or indirect knowledge of their needs, behaviors, and experiences is a long-standing challenge in HCI [23]. Built environments present a particularly demanding case. Physical spaces are shared by people with different goals, roles, memories, preferences, and capabilities. A first-time museum visitor navigating an unfamiliar atrium, a security guard completing a patrol route, and attendees gathering for an exhibition all occupy the same physical space, but all of these *building stakeholders*—people who occupy or use the space—experience it in substantially different ways. Because design decisions affect stakeholders differently, architects and other built-environment designers must integrate these different perspectives when evaluating their designs, often before a space physically exists and with limited direct access to future occupants.

Architects normally obtain stakeholder perspectives through client meetings, interviews, and other forms of user involvement. These practices provide valuable information, but identifying and accessing real future occupants can be a challenge. For example, a new laboratory in a university building may interface with an architectural firm through administrators, while the eventual occupants may primarily be students and professors. Even when direct stakeholders are accessible to the designer, opportunities for consultation are typically limited to particular stages of the design process and focus on hypothetical plans for a yet-unrealized space. Such constraints can make it difficult to gather realistic and repeated input from the right stakeholders about *spatial experience*, the subjective experience of an individual inhabiting a physical space.

Architects' current approaches towards these challenges include computational tools to evaluate yet-unbuilt spaces, such as Space Syntax for spatial configurations analysis [26], EnergyPlus for building performance simulation and energy modeling [16], and Pathfinder for occupant behavior modeling [58]. These examples demonstrate the value of

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such tools, but they primarily evaluate building and spatial performance, with limited representation of diverse future occupants or their varied spatial experiences.

Recent advances in large language models (LLMs) and generative agents, meanwhile, have demonstrated new possibilities for modeling people across different contexts, suggesting analogous opportunities for representing heterogeneous stakeholders in designed environments. LLM-based agents can incorporate different personas and contextual information while producing realistic and open-ended behaviors [45, 47]. Systems such as TravelAgent [44] and IndoorWorld [64] demonstrate potential for generative agents to interact with environments and produce information relevant to spatial evaluation. Most of this work, however, maps environmental information and agent personas directly to a single behavioral or affective output, leaving agents' internal processes of perception and reasoning—aspects crucially informative to architects and spatial designers—opaque.

We introduce the **Experience-Centered Spatial Simulation (ECSS) framework**, a computational framework for modeling stakeholder experience as an evolving interaction between environmental information and individual conditions. ECSS identifies the key states and relationships involved in spatial experience formation and organizes this process via three processors: the Sensory Processor, Environment Evaluator, and Personal Appraiser. By making these relationships computationally representable, ECSS supports transparent modeling of how different stakeholders experience the same environment differently, how the same individual experiences different environments, and how perception of one's environment evolves over time. To illustrate and evaluate the framework, we implement a computational instantiation in a sandbox environment and evaluate it with five controlled, proof-of-concept scenarios varying spatial layout, prior memory, stakeholder role, weather conditions, and occupancy.

This paper serves as a first step towards the use of generative AI agents for producing realistic, stakeholder-specific user experience feedback in yet-unbuilt architectural designs. To do so, it makes the following contributions:

(1) We introduce the Experience-Centered Spatial Simulation (ECSS) framework, a computational framework that represents spatial experience as an evolving interaction among environmental information, individual characteristics, and contextual states.
(2) We implement a simplified instantiation that operationalizes the framework using a web app simulator we developed for visualizing 2D architectural floor plans that generative AI agents can navigate and experience.
(3) We evaluate the framework through five proof-of-concept scenarios with our simulator and analyze the resulting environmental, experiential, and behavioral differences in generative AI agent behavior.

2 Related Work

We situate our work across four related fields: research on stakeholder differences in spatial experience, HCI work on user representations, simulation methods in architectural design, and emerging LLM-driven agents in spatial environments. These strands motivate our development of a computational framework for modeling spatial experience.

2.1 Spatial Experience and Stakeholder Differences

A space does not produce experience directly. Rather, its physical properties become meaningful in relation to the person who encounters them. Environmental psychology has long emphasized this reciprocal relationship between people and their surroundings [22, 56]. A space may have measurable properties such as geometry, illumination, or temperature, but experiential qualities such as comfort, or attractiveness arise as these conditions are interpreted by individuals. The same physical environment therefore can lead to different spatial experiences.

These differences are shaped by characteristics and situations of the people involved. For example, familiarity and prior knowledge can influence understanding and navigation of the same spatial setting [38], place evaluation reflects how well an environment supports specific goals [11], and individual physiology leads to wide variations in thermal comfort under identical conditions [37, 62]. These findings suggest that the meaning of an environmental condition depends jointly on what is present in the environment along with on who experiences it and under what circumstances.

This variation is particularly critical in architecture because built environments serve people with different capabilities, roles, knowledge, and needs. Inclusive and universal design philosophies emphasize the importance of considering diverse physical, sensory, cognitive, and behavioral needs [68]. Research on user involvement similarly highlights the value of stakeholders perspectives in the building design process. [9]. While post-occupancy evaluation (POE) and public-life observation methods could capture occupant perspectives after a building is inhabited [21, 41, 48], anticipating stakeholder-specific experiences during the design process remains a challenge [68, 69]. Because designers' own experiences cannot encompass the diversity of perspectives among future stakeholders, design requires ways to consider heterogeneous human experiences before a building is built. User representation offers one way to approach this challenge.

2.2 Representing Users in HCI

In HCI, researchers have long sought to represent users whose perspectives differ from those of designers. Approaches such as personas and scenario-based design provide structured ways to reason about users' profiles, goals, behaviors, and contexts during design [13, 23]. A related computational tradition represents users through explicit structures. Models such as GOMS decompose task performance into goals, operators, methods, and selection rules, while the Model Human Processor represents perceptual, cognitive, and motor processes to predict aspects of human performance [12]. Cognitive architectures such as ACT-R and EPIC similarly provide computational structures for modeling human cognition and behavior under different task conditions [3, 36]. Across these approaches, user representation provides a way to consider people whose perspectives and behaviors cannot be directly inferred from the designer's own experience.

Related efforts have emerged at the intersection of HCI and the built environment through Human-Building Interaction (HBI), an interdisciplinary area connecting HCI with architecture and urban design. HBI examines human experiences with and within built environments [1, 2], and prior work has brought HCI methods into architectural contexts [50, 59]. Modeling and simulation have been identified as one area where HBI can support architectural design and evaluation [7], but work in this area has largely addressed occupants in buildings that already exist, or evaluated design alternatives but in operational perspective [7, 53]. Representing occupants' experiences remains underexplored.

These traditions provide a basis for treating future stakeholder experience as a user-representation problem. Our work builds on this perspective by asking how heterogeneous spatial experiences can be computationally modeled during design.

2.3 Simulations in Architecture

Architecture has a parallel tradition of computational modeling and simulation to evaluate designs before construction. We organize these approaches into two categories by their primary modeling target: *building performance simulation* and *occupant simulation*. Building performance simulation focuses primarily on how buildings respond to environmental and operational conditions [17]. For example, EnergyPlus estimates energy use, heating and cooling loads, and thermal performance under different conditions [16]. Radiance models lighting conditions by simulating the distribution of

light within spaces [63], while computational fluid dynamics (CFD) focuses on airflow, ventilation, and wind around buildings [14].

Occupant simulations focus more directly on how people occupy, move through, and interact with buildings. They can represent occupancy, occupant locations, and movement trajectories [18], supporting analyses of circulation, route choice, congestion, and passenger flow. For example, queue-based models can quantify crowdedness [67], while Social Force Model represents pedestrian movement and interactions in crowd flow and evacuation [25]. A parallel line of work focuses on occupant behavior (OB), modeling how occupants adaptively interact with building systems through stochastic, statistical, and agent-based techniques [30, 65]. For instance, DNAs framework provides a structured representation of occupant actions and their relationships to building energy use [28, 29].

While powerful for evaluating environmental performance and observable behavior, these methods provide limited support for modeling how environmental conditions acquire different meanings across stakeholders. Individual differences are represented largely in service of performance-oriented objectives, and occupant models are applied predominantly during building operation rather than as design tools [4, 57]. Addressing this gap requires more flexible representations of individual profiles, histories, and context—capabilities that recent LLM-driven agents begin to provide.

2.4 LLM-driven Agents in Spatial Environments

Recent advances in LLMs have enabled generative agents that represent individuals through personas, memory, and context while producing open-ended, context-sensitive behavior. Park et al., for example, introduce a system architecture combining memory, reflection, and planning to present a simulated community [45], while the subsequent work extends generative agents to real-person profiles and large-scale social simulation with moderate accuracy [46]. Other work by Piao et al. propose AgentSociety, a large-scale simulation platform with a simulated social environment [47]. None of these efforts focus on architecture or design, but they demonstrate the potential of LLMs to support agents whose behavior evolves based on individual history and context over time.

Researchers has also extended LLM-driven agents into architectural settings. TravelAgent enables agents to navigate urban spaces using visual perception and chain-of-thought reasoning, producing spatial analyses [44]. IndoorWorld represents agents with differentiated roles, capabilities, and knowledge, and physiological needs in realistic office settings [64]. Other workd has used LLM-driven agents to generate occupant feedback based on real sensor data [40]. These studies suggest that LLM-based agents can incorporate increasingly rich individual and environmental information.

These advances create new opportunities for modeling stakeholders in built environments. However, existing LLM-driven agents typically map environmental observations and individual profiles directly to a single behavioral or affective output, treating spatial experience as a monolithic response rather than as the multi-dimensional and personalized process established above. This simplification makes the process of experience a black box, limiting the ability to explain *why* two stakeholders form different spatial experiences under the same environmental conditions, and in turn to relate those differences to specific design strategies. We therefore choose an explicit, staged computational structure for spatial experience formation in this work.

3 Modeling Commitments for Spatial Experience

This paper contributes a computational framework for modeling how spatial experience varies as different stakeholders (or agents) interact with the same environment, supporting generative agent-based space evaluations. Here, we define *spatial experience* as subjective experience of a physical space and *stakeholders* as different groups who may occupy or use the space. Rather than constructing a complete cognitive architecture or a psychological model, this framework

draws on perspectives from environmental psychology and spatial cognition to identify environmental and individual factors that are important to represent computationally.

A recurring perspective in environmental psychology is that people's experience of physical settings cannot be understood solely from properties of the environment itself, but emerges through the relationship among the person, environment, and their ongoing interaction [10, 32]. We translate this perspective into three modeling commitments: (1) *profile-sensitive*; (2) *time-responsive*, and (3) *environment-integrated*. These commitments define the minimal structure for the framework developed in this paper.

Profile sensitivity. Who is experiencing a space matters. Ecological and person–environment perspectives suggest that the significance of environment partly depends on individual characteristics. Lawton's ecological model emphasizes the relation between environmental demands and individual competence [39]. Kaplan and Kaplan further argue that environmental perception is shaped by prior experience [34]. Ishikawa and Montello further show persistent individual differences in integrating route knowledge [31]. These perspectives suggest that the same physical condition may not carry the same significance across individuals due to profile differences like perceptual capabilities or preferences. For example, a wheelchair user may attend more closely to accessible routes and spatial clearances than others; a person who prefers sunny, warm environments could place greater importance on windows or room orientation. The relevance of environmental features depends partly on the characteristics and state of the person experiencing them.

This leads to modeling commitment (1) *profile-sensitive*: a good model of spatial experience should capture differences across individual profiles and states. Rather than assuming a generic occupant, the model should allow individual attributes to influence the experience formation process.

Time responsiveness. Spatial experience is not a one-time response to a static environment; it evolves through continued perception and interaction over time. Neisser's perceptual cycle model describes perception as an ongoing relationship among existing knowledge, information seeking, action, and the environment, in which people actively seek information to guide their actions, while actions in turn influence their perceptions [43]. Classic accounts of spatial knowledge acquisition similarly propose that understanding of large-scale environments develops progressively through continued exploration and exposure [55]. As a result, the same person may generate different experience at different moments. For example, uncertainty in an unfamiliar corridor may decrease as recognizable landmarks or route familiarity develop. Although individual profile remains relatively stable, it may also change in the long term, such as mobility after an injury or preference shifts.

This motivates modeling commitment (2) *time-responsive*: spatial experience should be represented as a continually evolving process. The computational model should preserve and update relevant environmental and individual states across time, allowing prior interactions and current responses to influence subsequent experiences.

Environment integration. Finally, spatial experience is formed within a composite environment rather than a single dimension. Barker's ecological psychology treats behavior settings as organized relations between activities and their surrounding environmental milieu [5]. Canter similarly connects physical attributes with activities and people's understanding of the environment [10]. Computationally, this suggests that environmental representation should incorporate the different conditions relevant to a particular experience. However, existing architectural simulation tools often model factors such as geometry, daylight, or thermal conditions separately for domain-specific purposes. Our framework instead requires these factors to be considered together. For example, a large study area may look suitable

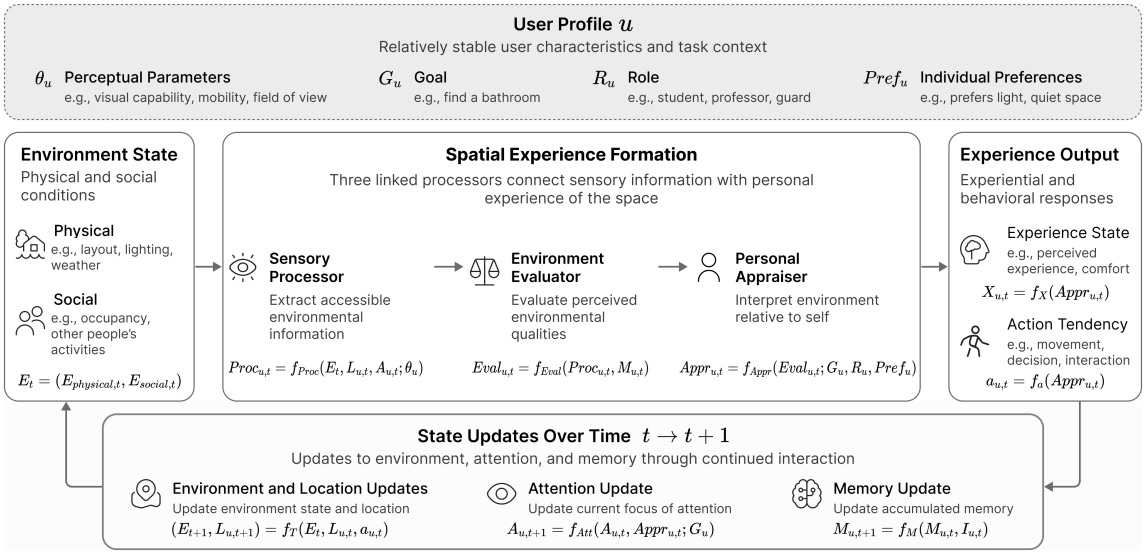


Fig. 1. The Experience-Centered Spatial Simulation framework models how different individuals experience the same physical environment. Environmental information is progressively perceived, evaluated, and appraised in relation to the individual profile, producing experience and action that update the simulation state across time.

for gathering from a geometric perspective, but excessive solar exposure or thermal discomfort may overshadow the physical layout in influencing how people experience and use the space.

Accordingly, our third modeling commitment (3) *environment-integrated* requires the joint representation of multiple relevant environmental conditions.

4 Experience-Centered Spatial Simulation Framework

In this section, we introduce the Experience-Centered Spatial Simulation (ECSS) framework. Building on the modeling commitments introduced in the previous section, we define its core components and formalize the process of spatial experience formation using mathematical representations.

4.1 Framework Overview

The ECSS framework represents how spatial experience forms through the interaction between environmental conditions and individual profiles over time. It is designed to support three properties: variation of individual profiles, continuous updating as people interact with an environment and integration of multiple environmental information.

ECSS decomposes spatial experience formation into three processing stages: the Sensory Processor, Environment Evaluator, and Personal Appraiser. Visualized in Figure 1 (and accompanied by Table 1 defining relevant variables), the framework takes inputs from the environment and progressively translates them into an individual's spatial experience. First, the environment is physically sensed, from these sensory inputs the user infers qualities of the environment, and then processes these according to her own preferences, goals, and other personal attributes. A subsequent Experience Generation stage translates the personal spatial experience into experiential and behavioral responses, a cycle that repeats at each time point.

Symbol	Description	Symbol	Description	Symbol	Description
a	Action tendency	L	Location	$Appr$	Personal appraisal
t	Current time step	M	Memory	$Eval$	Perceived environmental qualities
u	Individual	R	Role	$Pref$	Individual preferences
A	Attention state	X	Experience state	$Proc$	Perceptual evidence
E	Environment state	θ	Perceptual parameters	$f_{\{\text{Att},T,M\}}$	State-update functions
G	Goal	$\mathcal{I}$	Interaction information	$f_{\{a,X,\text{Appr,Eval,Proc}\}}$	Processing functions

Table 1. The above notation is used in our framework equations, and in Figure 1.

By separating spatial experience into this cyclical pipeline of components, the framework facilitates different computational implementations of each component. We describe our reference instantiation in Sec. 5. First, we describe each component in more detail below.

4.2 Individual and Environment State

The framework contains two complementary representations: the *environment state* and the *individual state*. The environment state describes environmental conditions that contribute to spatial experience, while the individual state represents the characteristics and current conditions of the person interacting with the environment. Together, they provide contextual information for spatial experience formation and are continuously updated over time.

4.2.1 Individual State. The framework illustrates representation of the individual interacting with the environment. For an individual u, this representation consists of both relatively stable *profile characteristics* and *dynamic states* that may change over time.

Profile characteristics describe relatively persistent differences between individuals that shape how environmental information is perceived. We represent these using four variables: goal G_u, role R_u, individual preference $Pref_u$, and perceptual parameters θ_u. G_u represents the individual's goal, such as finding a bathroom or a quiet place to work. R_u denotes the individual's role, which can further imply the individual's requirements or needs in a particular setting. Nurses in a hospital, for instance, prioritize efficient access to work areas, while patients care more about legibility. $Pref_u$ describes qualities that an individual u tends to favor or avoid, such as preferences for daylight, privacy, or quiet. θ_u captures the physical and perceptual characteristics that influence how the individual accesses and interacts with the environment. For example, an individual may have limited mobility, reduced vision, or higher sensitivity to particular conditions. These individual states do change over time, but at a slower cadence than each time point of the simulation (as with other variables we will see later).

In contrast to profile characteristics, dynamic states describe conditions that change as the individual interacts with the environment at each time step. At time step t, the current state includes the user's location $L_{u,t}$, representing where she is in the environment; attention state $A_{u,t}$, representing what aspects of the current environment are prioritized; memory $M_{u,t}$, representing her accumulated information or prior knowledge; and the evolving experience state, $X_{u,t}$, which summarizes the individual's experience up to the current point. These states are updated over time and influence subsequent cycles of experience formation. The resulting experience also produces an action tendency $a_{u,t}$, showing the individual's immediate inclination to respond to the current situation, such as continuing, pausing, changing direction or even more complicated behaviors.

4.2.2 Environment State. The environment state E_t represents information about the surrounding environment. At time step t, E_t is represented as a combination of physical and social environment state: $E_t = \left(E_{\text{physical},t}, E_{\text{social},t}\right)$. The physical component $E_{\text{physical},t}$ includes spatial geometry, lighting and acoustic conditions, materials, objects, signage, and other physical features of the environment. The social component $E_{\text{social},t}$ includes the presence, location, and activities of other occupants. This distinction is conceptual, as both environmental and social conditions jointly contribute to the perceptual evidence available to the individual at a particular situation. For example, occupants may obstruct sightlines, generate sound, restrict movement, or provide social cues about how a space is used.

Note that the framework does not prescribe a universal set of environmental variables. The environmental representation used in a particular instantiation depends on both the research question being investigated and the environmental information that can be reliably modeled or obtained. For example, an instantiation concerned with wayfinding can emphasize visibility, spatial configuration, and signage, whereas one concerned about working conditions may additionally incorporate daylight and acoustics. Likewise, the level of environmental detail is based on the available simulation methods and data sources. This means that environmental representation is extensible: different applications may integrate different combinations of environmental information according to their goals and technical capabilities.

4.3 Spatial Experience Formation

Spatial experience formation proceeds through three stages, operationalized by the Sensory Processor, the Environment Evaluator, and the Personal Appraiser. Together they transform current environment information into a personal appraisal of the environment to a particular individual.

4.3.1 Sensory Processor. This processor transforms the objective environment state into environmental information that individual users perceive. It addresses the question: **what environmental information is perceived by this individual?** Its output, the perceptual evidence available to individual u at time t, is defined as:

$$Proc_{u,t} = f_{\text{Proc}}(E_t, L_{u,t}, A_{u,t}; \theta_u). \tag{1}$$

The individual's location $L_{u,t}$, attention state $A_{u,t}$, and perceptual parameters θ_u jointly determine which parts of the environment are perceivable at a given moment. $L_{u,t}$ determines which parts of the environment are spatially accessible from the individual's current position and viewpoint, while $A_{u,t}$ determines which cues are prioritized. θ_u further modulates access to different sensory information according to individual capabilities. A guard may prioritize security-relevant cues, whereas a visitor may attend more to aesthetic features; similarly, a visually impaired individual may have limited access to visual cues, or a person with sharp hearing may be more sensitive to auditory information.

The resulting perceptual evidence $Proc_{u,t}$ represents the subset of the environment E_t that is accessible and attended to at time step t for individual u. Thus, two individuals in the same environment could consequently receive different perceptual evidence because they occupy different locations, orient toward different directions, allocate attention differently, or have different perceptual capabilities.

4.3.2 Environment Evaluator. Using sensory information from the previous step, the evaluator answers the question: **what qualities does the environment appear to have, given the current evidence and individual's prior spatial knowledge?** The perceived environmental qualities are represented as:

$$Eval_{u,t} = f_{\text{Eval}}(Proc_{u,t}, M_{u,t}) \tag{2}$$

Memory $M_{u,t}$ allows the current perceptual evidence to be interpreted in relation to prior encounters with the environment. This enables the same perceptual evidence to yield different environmental evaluations depending on what the individual already knows. For example, an empty corridor ambiguous to a first-time visitor would be comprehensible to a familiar occupant who knows where it leads.

Depending on the analytical context, the specific dimensions of $Eval_{u,t}$ may include qualities such as visibility, navigability, coherence, privacy, sensory intensity, or spatial enclosure. The framework does not prescribe a universal taxonomy of environmental qualities. Existing environmental psychology theories, architectural measurements, or task-specific criteria each could provide alternative operationalizations. The particular implementation used in our instantiation is described in Section 5.

4.3.3 Personal Appraiser. Using information from the previous step, the Personal Appraiser tries to answer the question: **what do these environmental qualities mean for this individual?** It does this by interpreting perceived environmental qualities according to the individual's current situation:

$$Appr_{u,t} = f_{\text{Appr}}(Eval_{u,t}; G_u, R_u, Pref_u) \tag{3}$$

While the Environment Evaluator evaluates how the environment appears, the Personal Appraiser relates those qualities to the individual's profile, including her relatively stable characteristics: G_u, R_u, and $Pref_u$. The presence of this processor means that even the same perceived environmental qualities can acquire different meanings. For example, an environment perceived as equally noisy and dim may be appraised differently by the same person during a social networking event compared to an online meeting.

4.4 Experience Generation

The Experience Generation stage externalizes the individual's spatial experience through two outputs: an experience state and an action tendency. These outputs are derived from the current personal appraisal:

$$X_{u,t} = f_X(Appr_{u,t}) \tag{4}$$

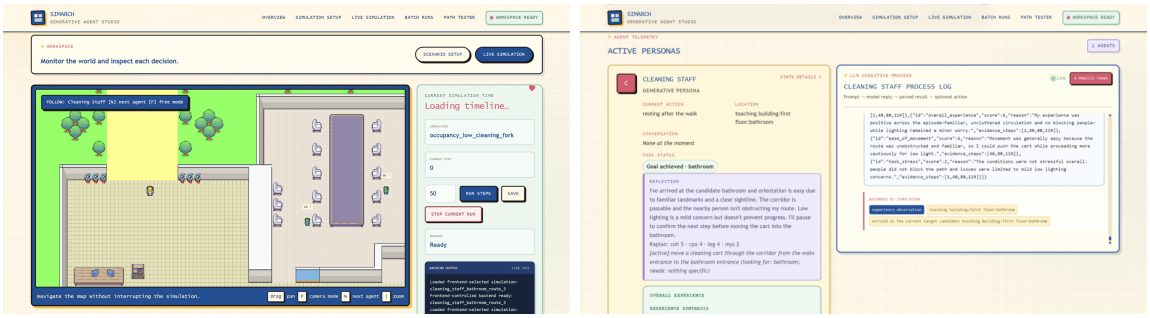
$$a_{u,t} = f_a(Appr_{u,t}) \tag{5}$$

$X_{u,t}$ summarizes the individual's experiential condition at the current point in the interaction, incorporating the prior context already reflected in $Appr_{u,t}$. $a_{u,t}$ captures the immediate behavioral inclination resulting from the appraisal and may influence subsequent interaction with the environment.

4.5 State Updates Over Time

In the real world, spatial experience evolves as individuals move through and interact with a space. To capture this dynamic, continual process, this framework models experience formation as an recursive iterative loop. At each cycle, the current individual and environmental states shape perception, evaluation, appraisal, and action. Actions and environmental dynamics then update the environment, location, attention, and memory for the next cycle.

4.5.1 Environment and Location Updates. The individual's action tendency can alter their position through actions such as moving, turning, waiting, or interacting with an environmental feature, thereby changing the information available at the following time step. Meanwhile, environment conditions, and the other occupants maybe also change as time progresses, causing both the physical and social environment to evolve. Together, these changes contribute to



(a) Simulation interface (b) Experience results interface

Fig. 2. The framework is instantiated through a web app simulator that visualizes the environment and agent interaction during a simulation run (a), while logging outputs and intermediate processing traces in parallel (b).

the updated environment state E_{t+1}:

$$(E_{t+1}, L_{u,t+1}) = f_T(E_t, L_{u,t}, a_{u,t}) \tag{6}$$

4.5.2 Attention Update. Attention is updated based on its previous state, the current appraisal, and the individual's goal. This captures both continuity in attention and appraisal-driven redirection. For example, an appraisal of low navigational control may redirect attention toward landmarks, or directional signage which could provide more evidence. We model this update as:

$$A_{u,t+1} = f_{\text{Att}}(A_{u,t}, Appr_{u,t}; G_u) \tag{7}$$

4.5.3 Memory Update. Memory evolves over time as the individual continues to interact with the environment, retaining and revising information accumulated from prior interactions. The update function f_M captures processes such as encoding, retention, revision, and forgetting. We denote by $\mathcal{I}_{u,t}$ the information from the current interaction that is available for memory encoding. The specific content of $\mathcal{I}_{u,t}$ and the mechanisms by which memory is updated vary across computational implementations; the framework requires only that relevant information from prior and current interactions can be carried forward to inform subsequent experience formation. We represent this process as:

$$M_{u,t+1} = f_M(M_{u,t}, \mathcal{I}_{u,t}) \tag{8}$$

5 Computational Instantiation

Having laid out this framework for representing individual-specific spatial experience through an iterative process of individual sensing, evaluation, personal appraisal, and actions over time, we implemented a simplified reference instantiation. Drawing on the simulation infrastructure and visual interaction pattern of Smallville [45], we developed a two-dimensional sandbox environment through a web app, and used GPT-5-mini to power a generative agent interacting with architectural floor plans. The instantiation implements a minimum set of capabilities required to realize the framework's end-to-end processing pipeline as a proof of concept.

5.1 Simplifying Assumptions

To make this first instantiation more tractable, we make three major simplifying assumptions during the implementation.

First, the built environment is represented as a simplified two-dimensional sandbox that includes only environmental dimensions relevant to the designed scenarios and that can be operationalized within the sandbox. This abstraction provides the minimum spatial structure needed to exercise the framework without reproducing the full complexity of a physical environment. This means, for instance, that agents in the space do not process z-axis dimensions like ceiling height or high-fidelity visual details.

Second, to focus on experiential differences arising from memories, roles, goals, and surrounding context, the instantiation assumes uniform perceptual capabilities across agents and does not model attention differences. Accordingly, θ_u is held constant, while A is omitted. Differences in agents' perceptual capabilities or selective attention are left for future implementations of the framework.

Third, the outputs produced by the reference instantiation are computational representations of the agent's simulated experience. They integrate evidence from simulated environmental conditions and the interaction process, such as low lighting levels, or crowded circulation, and provide interpretable feedback for comparing experiences across different simulated conditions. Since these representations are not calibrated against real human, their scores and descriptions should be interpreted comparatively within the simulation rather than as direct measurements of real-world human experience.

5.2 Component Implementation

5.2.1 Sensory Processor. This processor computes the environmental information directly available to the focal agent at a given moment. Under the assumptions described above, all agents share the same perceptual capabilities and attention configuration, so sensory state depends only on the current environment and the agent's location: $Proc_{u,t} = f_{\text{Proc}}(E_t, L_{u,t})$. Given these inputs, the processor computes the visible region and extracts the physical and social information available within that region.

Environmental conditions are represented through simple spatial models. For example, lighting and noise are generated from scenario-defined sources using spatial attenuation functions. Detailed formulations of these environmental calculations are provided in Appendix A.

5.2.2 Environmental Evaluator. In our reference instantiation, we implemented Environmental Evaluator with an LLM. The model receives the current perceptual evidence $Proc_{u,t}$ with memory $M_{u,t}$ and generates the perceived environmental qualities $Eval_{u,t}$. To provide a consistent structure for this aspect, we operationalize part of $Eval_{u,t}$ using four dimensions established by environmental psychologists Kaplan & Kaplan's environmental preference framework: coherence, complexity, legibility, and mystery (see Table 2 for detailed definitions) [34, 35]. The processor additionally summarizes scenario-relevant qualities such as openness, brightness, noisiness, and crowdedness. The resulting evaluation is then passed to the next processor. The prompt used in this step is shown in Appendix B.1.

Table 2. In our instantiation, the Environment Evaluator processor asks agents to evaluate the space on metrics defined by formative environmental psychologists Kaplan and Kaplan [34].

Dimension	Definition
Coherence	Perceived organization and consistency of the environment.
Complexity	Amount and variety of perceptual information in the environment.
Legibility	Ease of understanding spatial organization and maintaining orientation.
Mystery	Extent to which additional information is suggested beyond view.

5.2.3 Personal Appraiser. At each time step, the Personal Appraiser receives the perceived environmental qualities $Eval_{u,t}$, together with the individual's profile (G_u, R_u and $Pref_u$), and interaction context, such as current action and navigation state, and recent interaction history. The processor interprets the relevance of the perceived environmental qualities under these conditions and produces the personal appraisal $Appr_{u,t}$ using an LLM.

The same appraisal step also produces an immediate action tendency $a_{u,t}$ and auxiliary control signals that support subsequent interaction, including assessments of goal progress and whether the current action should be maintained or reconsidered. The implementation operationalizes these outputs through a structured schema, provided in Appendix B.2.

5.3 Simulation Procedure

For each simulation, we predefine the sandbox environment and scenario configuration, such as the spatial layout, agent's profile, starting location, etc. The simulator then initializes other occupants and their locations to represent the social environment. To reduce computational cost, the three processing components are not executed at every simulation time step. Instead, experience formation is triggered under three conditions: entering a new space, encountering a meaningful environmental change, or reaching a fixed number of time steps. This avoids repeated evaluation when the perceived conditions remain largely unchanged.

The simulation terminates when the agent completes its goal. At the end of each episode, a final experience summary is generated to synthesize the agent's experience. The prompt for this step is provided in Appendix B.3.

6 Comparative Scenarios

In this section, we design five comparative scenarios and use our reference instantiation to examine them. The goal is not to validate the accuracy of the framework as a predictive model, but to demonstrate its computational feasibility and its ability to capture experiential differences arising from changes in dimensions including individual profiles, environmental conditions, and interactions over time.

6.1 Scenario Design

To construct scenarios that are both empirically grounded and relevant to architectural design considerations, we focus on an academic building which serves diverse occupants, including students, faculty, staff, and visitors.

Using floor plans of a real campus building to which the author team had access, we constructed simplified representations of its second floor and designed five comparative scenarios with different experiential variations based on realistic use cases. One scenario focuses on the variation of the spatial configuration (layout), while the remaining scenarios instead vary memory, individual role, natural environment (weather) or social environment (occupancy) in a controlled manner. The scenarios therefore provide simple, complementary tests of the framework's ability to differentiate spatial experiences under different contextual conditions.

Layout Scenario. To examine how changes in spatial configuration influence the experience of the same agent performing the same task, we construct two simplified floor plans with different spatial relationships. In both conditions, the agent is a first-time visitor whose goal is to find a bathroom. As shown in Figure 3(a), the Direct layout provides a more direct path with greater visibility of the destination, whereas the Indirect layout introduces additional turns and a more ambiguous relationship between the primary circulation path and the destination. Other contextual conditions are kept consistent across the two conditions.

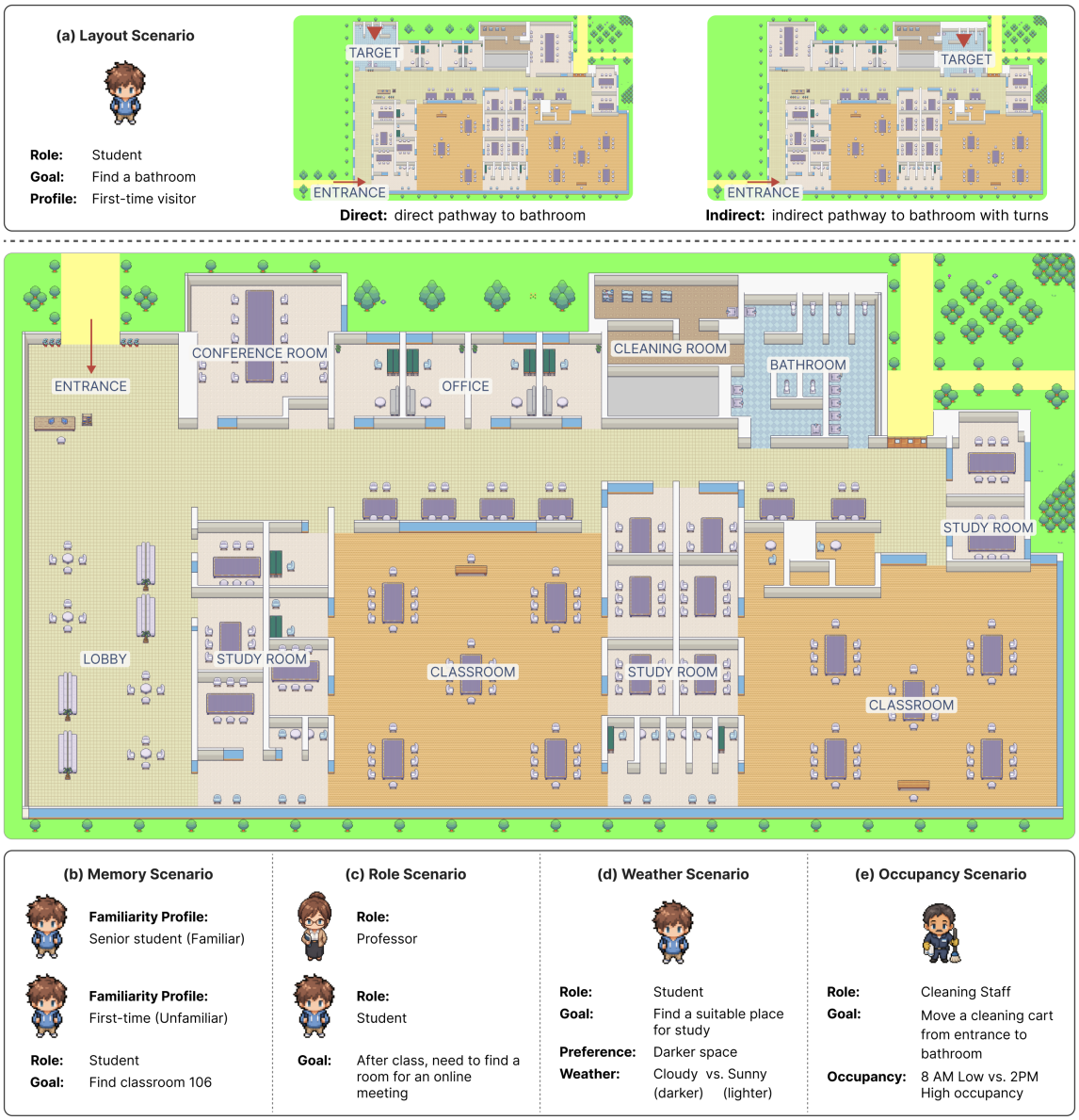


Fig. 3. Scenarios used in the experiments. Using a realistic academic building floor plan, we conducted controlled experiments varying layout (top), or attributes of the agent and environment (bottom) as illustrative proofs-of-concept.

Memory Scenario. This scenario, shown in Figure 3(b), focuses on memory. We compare a new unfamiliar student with a senior student who is familiar with the space. Both their goals are to find classroom 106, but the former agent has no memory of the space while the latter has been to this classroom before. All other aspects of the simulation are held constant, to demonstrate how differences in memory or accumulated spatial knowledge influence the interpretation of the environment.

Measure	Definition	Scenario(s)
Experience Metrics		
Goal Support	Extent to which the environment supports the current goal	1–5
Overall Experience	Overall experience during the entire process	1–5
Navigation Confidence	Confidence in knowing where to go	1, 2
Route Clarity	Clarity of the spatial layout to find the route	1
Navigation Control	Perceived control over the navigation process	2
Task Control	Perceived ability to carry out the intended activity	3
Task Suitability	Suitability of the environment for the intended activity	3
Visual Comfort	Comfort under the current visual conditions	4
Preference Match	Match between environmental conditions and individual preference	4
Ease of Movement	Ease of moving through the environment	5
Task Stress (reverse-scored)	Stress when performing the current task; higher values indicate lower stress	5
Simulation Process Metrics		
Steps to Completion	Number of simulation steps until completion	1–5
Candidates Rejected	Number of candidate options rejected	1–5
Path Efficiency	Shortest-path movement steps ÷ Current path movement steps	1–5

Table 3. Each of our five proof-of-concept scenarios are evaluated on four Experience Metrics and three Simulation Process Metrics, labeled above. Scenario numbers correspond to: (1) Layout, (2) Memory, (3) Role, (4) Weather, and (5) Occupancy.

Role Scenario. In the role scenario, environmental variables are held constant while the agent's role is varied. In one condition the agent's role is Student, while in the other it is Professor. As illustrated in Figure 3(c), both agents are leaving the same classroom and trying to find a space for an online meeting. Relevant to this goal, the professor has an office, while the student does not.

Weather Scenario. This scenario investigates the impact of natural environmental conditions on spatial experiences. Figure 3(d) illustrates a student who prefers darker spaces and is seeking a comfortable place to study under two weather conditions. Cloudy weather provides lower levels of natural light, while sunny weather brings greater daylight into an otherwise-identical setting.

Occupancy Scenario. The last scenario explores how social environment alters the experience of the same individual performing the same task. In Figure 3(e), a cleaning staff member moves a cleaning cart from the entrance to the bathroom under different occupancy levels: low occupancy at 8 AM and high occupancy at 2 PM. The spatial configuration, personal profile, and route remain unchanged.

6.2 Analysis Method

Each condition for each scenario was repeated ten times. Within each scenario, only the factor of interest was varied, while other parameters were controlled. Runs were generated independently, so each run across conditions was treated as an independent test.

Our quantitative analysis mainly considered two groups of measures: *experience metrics* and *task process metrics*. For experience metrics, each completed episode was assessed using four predefined indicators on 7-point Likert scale, as answered by the given agent when prompted at the end of the run. For precise prompts, see Appendix B.4. Two indicators, *goal support* and *overall experience*, were shared across all scenarios, while two additional indicators were defined based on the experiential variation targeted by the given scenario. Meanwhile, the *task process metrics* characterized how the

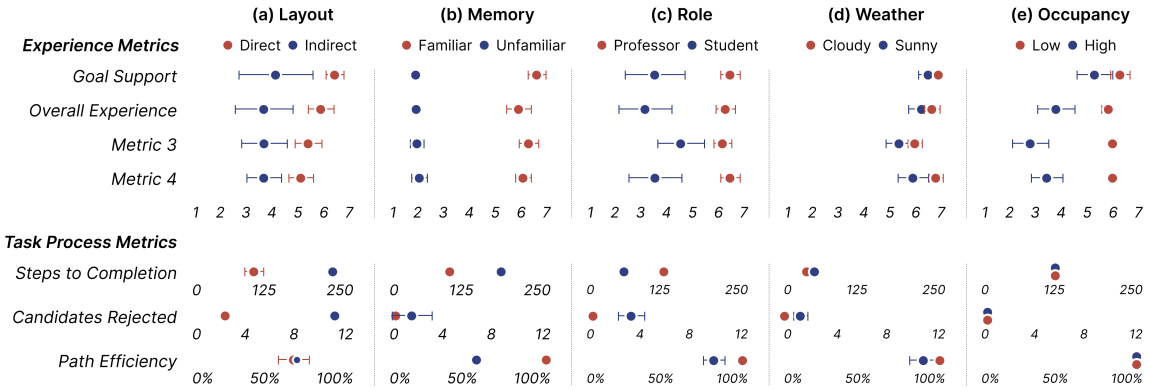


Fig. 4. Mean scores (±1 SD) on the primary metrics across conditions for all scenarios (10 runs each). The first two experience metrics, *Goal Support* and *Overall Experience*, are common across scenarios, whereas the third and fourth are scenario-specific and detailed in Table 3. Across most primary metrics, the compared conditions show clear and directionally consistent separation.

simulated individual progresses through the assigned task, including *steps to completion*, *candidates rejected* and *path efficiency*(shortest/actual). The complete set of measures and their definitions is provided in Table 3.

For each measure, we calculated the mean (M) and standard deviation (SD) for each condition, the between-condition mean difference (MD) with a 95% bootstrap confidence interval (CI), and results from two-sided Welch's t-tests and independent permutation tests. Welch's t-tests served as the primary inferential tests, with Holm–Bonferroni corrections applied separately within the experience and task process measure families for each scenario [27]. Statistical significance was determined using the adjusted p-values at $\alpha = .05$. Independent permutation tests were used as robustness checks.

In addition to the primary outcome measures, Kaplan-based outputs from the Environment Evaluator were examined as intermediate measures, but were not included in the primary inferential analysis. Representative processing traces and generated experiential descriptions were also reported as qualitative results to illustrate how the observed quantitative differences emerge.

6.3 Scenario Results

In this section, we summarize the results of the five diagnostic scenarios and analyze how the framework differentiates experiences across dimensions of interest. Figure 4 illustrates the primary measures for all scenarios using mean and standard deviation, showing distinct experiential patterns between conditions. Complete descriptive statistics, together with the intermediate Kaplan-based evaluations, are provided in Appendix C.

6.3.1 Layout Scenario. We used *navigation confidence* and *route clarity* as additional experience metrics 3 and 4. Compared with Indirect layout, Direct layout produced consistently more positive experience ratings, with higher goal support ($M = 6.50$ vs. 4.20, $p_{adj} = .005$), overall experience ($M = 6.00$ vs. 3.80, $p_{adj} = .004$), navigation confidence ($M = 5.50$ vs. 3.80, $p_{adj} = .004$), and route clarity ($M = 5.20$ vs. 3.80, $p_{adj} = .004$). Independent permutation tests yielded the same significance pattern.

The task-process measures distinguish absolute navigation burden from relative path efficiency. Agents in Direct layout completed the task in fewer steps and rejected fewer candidate spaces than those in Indirect layout (both $p_{adj} < .001$). However, path efficiency was nearly identical across layouts ($M = 0.65$ vs. 0.66, $p_{adj} = .829$). This is because

agents entered both layouts as first-time visitors without prior spatial memory, exploratory wayfinding remained evident in both conditions. For instance, agent `A-direct-02`[1] generated in qualitative feedback: "I inspected classroom 109 and study rooms before committing to the nearest visible candidate, and ultimately confirmed the bathroom after some detours." Similarly, `A-indirect-07` describes the process as "repeatedly inspected many similar rooms and offices" but "finally found the bathroom".

Overall, the results show that the framework can translate differences in spatial configuration into corresponding differences in simulated experience. A more direct route to the bathroom supported more positive experience ratings and lower search burden, whereas the less direct layout required more effort.

6.3.2 Memory Scenario. The additional experience metrics used in memory scenario were *navigation confidence* and *navigation control*. Compared with the Unfamiliar condition, the Familiar condition showed substantially higher goal support (M = 6.70 vs. 2.00), overall experience (M = 6.00 vs. 2.00), navigation confidence (M = 6.40 vs. 2.10), and navigation control (M = 6.20 vs. 2.20). All four differences remained significant after Holm correction ($p_{\text{adj}} < .001$) with permutation tests yielding consistent results. The navigation process showed a corresponding pattern. Familiar agents completed the task in 94 steps, compared with 177 steps in the Unfamiliar runs, and followed the shortest feasible route (M = 100% vs. 55% in path efficiency). Unfamiliar agents also rejected an average of 1.40 spaces before reaching the target, whereas Familiar agents navigated directly to the remembered destination.

The generated experiential descriptions illustrate how these differences showed up in individual simulations. `A-familiar-03` rated its navigation confidence as 7 and explained, "I felt very confident about where to go because remembered landmarks and clear sightlines kept me on a committed route toward room 106." In contrast, `A-unfamiliar-04`, which rated navigation confidence as 2, reported, "I was often uncertain about where to go (many checkpoints marked 'uncertain' and recommendations to request a new decision)."

The results of both experiential and process outcomes suggest that the framework is able to differentiate stakeholders with different levels of memories in their experiences and navigation of the same environment.

6.3.3 Role Scenario. In the role scenario, where Professor and Student agents needed to find an online meeting space, our additional experiential metrics were task control and task suitability. Professor agents showed higher ratings on all dimensions: goal support (M = 6.60 vs. 3.70, $p_{\text{adj}} < .001$), overall experience (M = 6.40 vs. 3.30, $p_{\text{adj}} < .001$), task control (M = 6.30 vs. 4.70, $p_{\text{adj}} = .003$), and task suitability (M = 6.60 vs. 3.70, $p_{\text{adj}} < .001$).

"Noise" and "crowd" appeared repeatedly as keywords in the generated feedback from Students, who also frequently described some spaces as "inappropriate" fits. Although the average steps to completion for Students were less than Professors', the average path efficiency (M = 81.50%) was lower, and the number of candidates rejected (M = 3.40) was much higher. Students needed to check three rooms on average to find a suitable space while Professor agents consistently returned to their offices with zero rejected candidates and path efficiency of 100%. As `A-professor-02` stated, "confident that the environment was coherent and legible, I maintained a committed route and continued directly toward my office. Nearby people and furnishings were noticeable but did not obstruct my passage, so I proceeded without hesitation."

Even though the agents' goals were the same in this task, Students and Professors had different experiences because of their different roles, demonstrating that ECSS can successfully reflected how roles influence spatial experience even when individuals have the same task demands.

[1] `A-<condition>-<run number>` denotes a simulated agent under the specified scenario condition and experimental run.

6.3.4 Weather Scenario. To understand how weather conditions can influence users' experiences based on their preferences, we used *visual comfort* and *preference match* as additional experience metrics. For an agent who preferred darker spaces and sought a comfortable place to study, cloudy conditions produced slightly higher scores across all experiential measures. The mean differences in goal support, overall experience, and visual comfort were $MD = 0.40$, 0.40, and 0.60, respectively, although they were not statistically significant. Preference match showed the clearest difference, with higher ratings under Cloudy conditions ($MD = 0.90$, $p_{\text{adj}} = .029$).

Agents in the Sunny condition also required more effort to find suitable rooms, exploring more spaces ($M = 1.00$ vs. 0.00) with lower efficiency ($M = 89\%$ vs. 100%) and more steps ($M = 50.20$ vs. 38.80). `A-sunny-07` described that "lighting was often brighter than my preference (leading to two rejections)," but continued, "I was able to locate a dimmer seat in the final room, giving moderate visual comfort." In contrast, `A-cloudy-08` quickly concluded after entering one room, "the room consistently provided table/chair and darker, low-noise quarters," allowing the agent to settle in study room 105.

This scenario illustrates how the same spatial configurations produce different experiences on selected dimensions when environmental conditions interact with individual preferences.

6.3.5 Occupancy Scenario. We used *ease of movement*, and *task stress* (reversed in Figure 4) as additional experience metrics to evaluate the impact of occupancy. Overall, simulations in the Low Occupancy condition produced significantly lower task stress ($MD = -2.50$, $p_{\text{adj}} < .001$) and significantly higher scores on goal support($MD = 1.00$, $p_{\text{adj}} = .012$), overall experience ($MD = 2.00$, $p_{\text{adj}} < .001$), and ease of movement($MD = 3.10$, $p_{\text{adj}} < .001$), aligning with expected directions. The largest difference appeared in ease of movement. Nine of the ten simulated cleaning staff agents in the High Occupancy condition described the cleaning cart as making movement more difficult. For example, `A-high-10` reported that "a high visible people count, ambient noise, and dim lighting made pushing the cleaning cart awkward." Similarly, `A-high-01` noted that the environment "made me pause and reassess whether it was safe to push the cleaning cart."

At the same time, goal support remained relatively high in both conditions. `A-low-04` rated goal support as 7 because of "strong legibility, remembered connections, minimal obstructions," while `A-high-09`, which rated goal support 5 out of 7, noted that "legibility and my memory made the bathroom easy to locate." These comments suggested that the clear and legible corridor layout continued to support task completion even as crowding made movement more difficult.

The process measures were identical because of the task design. But the experiential differences emerged despite following the same route. This scenario confirmed that the framework can represent other occupants as part of the social environment, and capture how occupancy influences the spatial experience.

7 Discussion

In the previous sections, we introduced ECSS, a computational framework for modeling how different stakeholders experience the same spatial environments. Through an initial implementation and five comparative scenarios, we showed that the framework can produce distinguishable differences. These results demonstrate the potential of making heterogeneous spatial experiences computationally explicit, but raise further questions about how such simulations should be developed and interpreted. In this section, we reflect on these implications, including limitations and future work, from three perspectives: the challenges of operationalization, the scope and generalizability of the framework, and the ethical considerations of synthetic stakeholder simulation in this context.

7.1 Challenges in Operationalization

7.1.1 Grounding and Fidelity. The interpretable and statistically significant differences observed across most scenarios demonstrate the ECSS framework's ability to capture variations across spatial, environmental, and stakeholder conditions. However, these results are initial, small-scale tests and do not establish fidelity to real human experience. Notably, future evaluation of a rigorous ECSS system will need to be calibrated to and validated against human-reported feedback.

A challenge in building a more robust system is how faithfully the underlying interaction processes can be operationalized. Wayfinding provides a useful example. Human navigation involves perception, prior knowledge, attention, spatial learning, uncertainty, and movement, many of which aspects are omitted or simplified in our current implementation. In this work, we use breadth-first search (BFS) for route finding and simple rules for selecting candidate destinations. We experimented with more direct LLM-based spatial reasoning, including using 2D screenshots and symbolic maps as input for spatial understanding, but found that model interpretation was too sensitive to the representation style and did not always preserve the intended spatial structure. Thus despite our initial positive results, we anticipate wayfinding, and other low-level interaction processes, will pose a substantial modeling challenge and need to be built into the framework as more sophisticated components.

Similar trade-offs apply to the model's representation of the physical environment. The current prototype models only the select spatial and environmental conditions needed for our comparative scenarios. Fundamentally, the environmental information available to the simulation is constrained by two main factors: the available data, and the computational methods needed to interpret it. Richer forms of environmental information could therefore be incorporated through corresponding processing components. For example, 3D geometry or egocentric visual observations could be processed through visual perception components. Acoustics, thermal conditions, smell, material qualities, or dynamic social activity could also be incorporated through specialized sensing or inference components. Systems like SayNav illustrate how richer spatial and multimodal representations can support agent interaction in simulated environments [49]. Future work could combine such representations with empirically grounded behavioral models and calibration against human behavioral and self-report data. This would help determine which simulated responses reliably correspond to human experience and which are better treated as exploratory signals for design.

7.1.2 From Differentiation to Design Utility. Beyond improving the fidelity of the simulation, another important question is whether the resulting representations of spatial experience can meaningfully support architectural design. A central motivation for this work is to bring more stakeholder perspectives into the architectural design process. But even a framework fidelitous to such experiences will need itself to be thoughtfully designed and presented to be useful to architects. In particular, the qualitative descriptions generated in our tests often focused on experiential concerns (such as unclear navigation, crowding, or mismatched environmental conditions), rather than providing specific design actions that architects might prefer.

So how should simulated experience be communicated to designers? Architectural simulation tools often communicate their results through visualizations, which align with architects' existing representational conventions. For example, tools for circulation simulation often use trajectories, density maps, and animations to communicate circulation patterns. A similar approach could be explored for experiential simulation. Environmental conditions and simulated experiential responses could be presented through spatial visualizations, comparative summaries, or qualitative explanations that allow designers to identify where and why different stakeholders have different experiences.

The usefulness of feedback also depends on how it fits into existing architectural workflows. Architects normally have their design process to work with their clients. Future work is needed to examine where experience-centered simulation

could contribute within these practices, what forms of feedback architects find actionable, and how much additional effort they are willing to incorporate into their existing process. Interviews or co-design studies with practicing architects across different firms could help identify their needs and inform the development of future architect-facing tools based on the framework.

7.2 Scope and Generalizability of the Framework

7.2.1 One Way Among Many. A central modeling choice in this work is to decompose the experience-formation process into a series of explicit stages, rather than relying on a direct, end-to-end mapping from environmental and individual information straight to experiential output. This decomposition makes intermediate representations available for inspection and allows individual components or inputs to be varied independently. For example, a person with high sensitivity to light and one who simply prefers darker spaces may both experience discomfort in a bright environment, but for different reasons; the framework makes different pathways examinable. We believe making such distinctions explicit is particularly useful for architectural design, which cares about the *why* behind an individual's experience.

Of course, the use of generative models means that even an explicit framework does not make for a fully explainable model. The internal reasoning of an LLM remains largely opaque. What ECSS makes inspectable is the computational structure around that reasoning: what information enters each component, what intermediate states are, and how changes in different factors are reflected in the corresponding simulated experience. The value of this decomposition is therefore not in establishing a single correct methodology of experience formation, but in providing a clear and executable structure that can be tested, revised, and compared with alternatives.

More fundamentally, ECSS should be understood as just one possible computational decomposition of spatial experience rather than a definitive model of human cognition. It is informed by prior work on person–environment interaction, environmental psychology, and spatial cognition, but the particular organization of environment perception, evaluation, personal appraisal is our invention. Human experience in physical environments is more complex, and other frameworks could identify different components, relationships, or levels of abstraction.

7.2.2 Extending ECSS Within and Beyond Architecture. The framework defines a minimal set of constructs to differentiate spatial experiences and make them computationally explicit, but is not an exhaustive representation of the individual or environment. For example, stakeholder profiles in our current formulation include perceptual parameters, goal, role, and preferences, but additional characteristics could be introduced when relevant to a particular application, such as personal background or domain-specific knowledge. Memory representations, and experience outputs could likewise be extended or specialized under the framework.

ECSS could also be extended for use beyond architectural design. Existing work has explored LLM-based and generative agents in virtual worlds, games, and embodied-agent simulations, with particular emphasis on memory, planning, interaction, navigation, and task completion. LLM-driven NPCs have been studied in game environments such as Minecraft [51]. Embodied-agent systems such as Voyager and SimWorld further show how agents can act over time in interactive environments using accumulated experience and spatial information [61, 66].

Our framework could provide agents with a structured mechanism for perceiving and understanding environment. Rather than representing heterogeneity only through different personas or instructions, agents could behave differently based on their profiles, available environmental information, prior knowledge, and memories. In a gaming context, this could be useful for powering non-player characters (NPCs), where richer differences in perception and interpretation could support more believable interactions and more diverse player experiences. In embodied-agent and virtual

world systems, ECSS could help structure how agents continuously sense, evaluate, and respond to the surrounding environment to support more context-sensitive agent–environment interaction. We see many possible opportunities to evaluate ECSS in, or extend it for, other settings.

7.3 Simulated Stakeholders as Augmentative Design Support

7.3.1 Limits and Risks of Synthetic Stakeholder Simulation. Along with the increasing use of LLM simulation and synthetic users, a growing body of work has questioned whether LLM-generated agents can serve as reliable proxies for real human subjects. One common critique concerns diversity: although synthetic responses appear plausible, they may homogenize perspectives and flatten variation [8, 20]. These concerns become more consequential when agents are conditioned on demographic identities or persona descriptions. Identity-based prompting can essentialize social groups, while biases in the underlying model may reproduce distorted representations, particularly for marginalized populations [15, 60]. Persona-based their limited transparency and reproducibility [6], and for sometimes degrading model performance [24].

More fundamentally, simulating a stakeholder is not equivalent to involving that stakeholder. Prior critiques argue that synthetic participants lack the palpability, agency, and situated knowledge that make human participation meaningful, and that their use may erase rather than represent the communities they stand in for [33]. From a participatory design perspective, even with perfect fidelity, synthetic agents offer a resemblance of participation rather than participation itself [42]. These critiques reflect a broader limitation of synthetic-user simulation: a computational persona can easily be granted representational authority beyond what it can substantiate. The challenge is not only whether a simulation is valid or plausible, but also what authority should be assigned to it as a representation of a person or a stakeholder group.

7.3.2 Positioning ECSS: Rehearsal, Not Substitution. These limitations do not eliminate all potential value of simulation, but they do constrain the role it should play. Architecture provides a particular opportunity because many design decisions must be made before the space can be experienced by future occupants. Designers can and do consult clients, occupants, and communities during the design process, but these engagements typically occur at snapshot moments, rather than continually over time. Further, the client, the individual(s) who hired the designer and with whom that designer can most easily communicate, is not always the eventual user of the space. Throughout this process, designers need to continually make decisions that consider how a wide range of stakeholders will experience the space.

We therefore view ECSS and systems built with it as a form of design rehearsal, not a substitute for human participation. By instantiating different stakeholders' profiles, designers could use the framework to explore how design might impact their experiences before construction. The value of such simulation is not about predicting what an actual stakeholder "would say", but uncovering possible design concerns, and revealing assumptions that may warrant further investigation. Simulation can therefore serve as a complementary form of design support rather than as an authoritative representation of stakeholder groups. Importantly, simulation should not be isolated from real stakeholder engagement. Future systems could use interviews, participatory activities, or other empirical evidence to inform and calibrate the profiles and assumptions used in simulations, and simulations could be used between rounds of real human engagement. In this way, we see potential for ECSS to augment human-centered design processes rather than replace or undermine them.

This positioning is consistent with the broader tradition of intelligence augmentation and human-centered AI, which frames computational systems as tools for supporting and extending human reasoning while preserving human

judgment [19, 52, 54]. Under this positioning, the contribution of synthetic simulation is not to speak on behalf of stakeholders, but to provide designers with a computational tool for considering stakeholder experiences throughout the design process.

8 Conclusion

This paper presents Experience-Centered Spatial Simulation (ECSS), a computational framework for modeling stakeholders' spatial experiences. By decomposing experience formation into explicit components that connect environmental information and individual characteristics, ECSS provides a structured method to make heterogeneous spatial experiences computationally representable and inspectable. We implemented an instantiation of the framework using generative AI agents in the context of architectural design evaluations, and tested five comparative scenarios. Our results find that the framework does produce different differentiate experiential outcomes across factors of interest, and provide an initial demonstration of experience-centered spatial simulation as a computational approach for architectural design. We also lay out directions for future work; realizing the framework's value beyond proof-of-concept examples will require validation against human experience, and tailoring to architectural designers' workflows. We position ECSS not as a substitute for stakeholder participation or designer judgment, but as an augmentation tool helping designers surface possible experiential differences among stakeholders.

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A Environmental Field Computation

The reference instantiation computes several local environmental signals that are provided to the Sensory Processor. These signals are normalized representations used to introduce controlled spatial and temporal variation into the simulation rather than physically calibrated simulation of real spaces. We model three environmental dimensions: lighting, noise-related activity, and perceptual openness.

A.1 Lighting Field

Lighting combines artificial lights and daylight entering through windows. The lighting signal Λ_t at the focal individual's location is:

$$\begin{aligned}\Lambda_t = \mathrm{clip}_{[0,1]}\Bigg(&\sum_{s\in\mathcal{S}_\Lambda} I_s^\Lambda \left[1 - \frac{d(L_s, L_{u,t})}{R_s^\Lambda}\right]_+ T_\Lambda(L_s, L_{u,t}) \\ &+ \omega f_{\mathrm{day}}(h_t)\left[\min\left(\sum_{\mathbf{w}\in\mathcal{W}} I_W^\Lambda \left[1 - \frac{d(\mathbf{w}, L_{u,t})}{R_W^\Lambda}\right]_+ T_\Lambda(\mathbf{w}, L_{u,t}),\ \Lambda_{\max}^W\right) + \Lambda_{\mathrm{amb}}\right]\Bigg),\end{aligned} \tag{9}$$

where $[u]_+ = \max(0, u)$, $d(\cdot,\cdot)$ denotes Euclidean distance, $\mathcal{S}_\Lambda$ is the set of static artificial light sources, $\mathcal{W}$ is the set of window cells, Λ_{amb} is a constant indoor ambient daylight term ($\Lambda_{\max}^W = 1$ and $\Lambda_{\mathrm{amb}} = 0.25$ in the reference instantiation). Material-dependent transmission is defined as:

$$T_\Lambda(L_s, L_{u,t}) = \prod_{\mathbf{z}\in\mathcal{R}(L_s, L_{u,t})} \tau_\Lambda(\mathbf{z}), \tag{10}$$

where $\mathcal{R}(L_s, L_{u,t})$ contains the intermediate grid cells intersected by the source–target ray and $\tau_\Lambda(\mathbf{z}) \in [0,1]$ is the corresponding light-transmission coefficient.

The weather multiplier $\omega \in [0,1]$ controls scenario-level daylight intensity, while $f_{\mathrm{day}}(h_t) \in [0,1]$ represents time-of-day variation. In the implementation, f_{day} increases linearly from sunrise to full daylight, remains constant during daytime, and decreases linearly toward sunset.

A.2 Acoustic Field

Ambient and agent-generated sound is propagated over the four-neighbour grid, allowing sound to travel through openings and around corners. For a source s, the physical sound level at the focal individual's location is determined by the highest-transmission path between source and target:

$$N_s^{\mathrm{phys}}(L_{u,t}) = \max_{\pi\in\mathcal{P}(L_s, L_{u,t})} \left[I_s^N \prod_{k=1}^{K} \alpha_N \tau_N(\mathbf{z}_k)\right], \tag{11}$$

where $\pi = (\mathbf{z}_0, \ldots, \mathbf{z}_K)$ is a valid grid path from $\mathbf{z}_0 = L_s$ to $\mathbf{z}_K = L_{u,t}$, $\alpha_N \in (0,1)$ is the per-tile retention factor, and $\tau_N(\mathbf{z}_k) \in [0,1]$ is the material-dependent sound-transmission coefficient.

The field is implemented using a priority-queue flood and truncated when the propagated level falls below a small numerical threshold. Contributions from scenario-defined ambient sources and other active simulated agents are additive:

$$N_{\mathrm{phys}}(L_{u,t}, t) = \sum_{s\in\mathcal{S}_N} N_s^{\mathrm{phys}}(L_{u,t}) + \sum_{a\in\mathcal{A}_t} N_a^{\mathrm{phys}}(L_{u,t}, t), \tag{12}$$

where $\mathcal{S}_N$ denotes static ambient sound sources and $\mathcal{A}_t$ denotes other active simulated agents.

A.3 Perceived Social Activity

Lightweight scenario actors that are not modeled as full acoustic sources contribute an additional local proxy for perceived social activity. For visible actor a, this contribution decreases linearly with distance. The aggregate contribution from visible actors is:

$$N_{\mathrm{soc},t} = \sum_{a \in \mathcal{V}_t} q_a \left[1 - \frac{\min(d(L_{a,t}, L_{u,t}), R_{\mathrm{soc}})}{R_{\mathrm{soc}}} \right]_+, \tag{13}$$

where $\mathcal{V}_t$ is the set of scenario actors currently visible to the focal individual, q_a is the actor's scenario-defined activity-noise value, and R_{soc} is the effective radius of the social contribution.

The resulting noise-related perceptual signal is:

$$N_t = \mathrm{clip}_{[0,1]} \left(N_{\mathrm{phys}}(L_{u,t}, t) + N_{\mathrm{soc},t} \right). \tag{14}$$

The social term should be interpreted as a proxy for locally perceived social activity rather than as a physically complete acoustic simulation.

A.4 Perceptual Openness

Perceptual openness is derived from the amount of walkable area contained in the focal individual's unobstructed visual field. Let $Y_{\mathrm{vis}}(L_{u,t})$ denote the number of visible walkable cells. Openness is computed as:

$$O_t = 1 - \exp\left(-\frac{Y_{\mathrm{vis}}(L_{u,t})}{Y_0} \right), \tag{15}$$

where Y_0 controls the rate at which the measure approaches its upper bound.

A.5 Environmental Vector

The three normalized environmental signals are combined into a local environmental vector:

$$\mathbf{e}_t = \begin{bmatrix} \Lambda_t & N_t & O_t \end{bmatrix}^{\mathsf{T}}. \tag{16}$$

B Prompt Templates

B.1 Environment Evaluator

The Environment Evaluator receives the current perceived environmental information and recent spatial memory. The complete prompt template is shown below.

```
You are the Environment Evaluator in a spatial-experience simulation process. Evaluate what the
currently perceived environment is like, and how its perceived organization relates to recent spatial memory.

Current sensed environment: !<INPUT 0>!
Recent environmental memory: !<INPUT 1>!
```

```
Tasks:
1. Score Kaplan coherence, complexity, legibility, and mystery from 1 to 5.
2. Give one concrete evidence sentence for each score.
3. Cite the supplied input fields used for each score.
4. Compare the current perception with recent environmental memory when meaningful.
5. Summarize the currently perceived environmental qualities.

Evaluation guidance:
- coherence:
  Evaluate how organized, unified, and structurally consistent the currently perceived environment appears.
  1 = fragmented or disordered, 3 = partly organized, 5 = strongly unified

- complexity:
  Evaluate the amount and variety of currently perceptible spatial and environmental information.
  1 = sparse or monotonous, 3 = moderate variety, 5 = rich but still perceptible

- legibility:
  Evaluate how clearly the current spatial organization supports orientation and understanding of spatial
  relationships.
  1 = difficult to understand or orient, 3 = mixed orientation cues, 5 = clear organization, landmarks, and
      route structure

  Recent memory may strengthen legibility when repeated landmarks, remembered route relations, or consistent
  spatial structure provide additional orientation support.

- mystery:
  Evaluate the extent to which the current view suggests additional spatial information beyond what is
      immediately visible.
  1 = little indication of further information, 3 = some partially revealed extension, 5 = strong indication
      of additional information

Memory and temporal comparison:
- Treat the current sensed environment as the primary evidence for the current
  evaluation.
- Use recent environmental memory as temporal and orientation context.
- Compare current and remembered observations when they reveal meaningful
  continuity or change in spatial organization, landmarks, openings,
  visibility, or orientation support.
- Memory may support perceived legibility by connecting current observations
  with previously encountered spatial information.
- Do not add previously perceived but currently unseen objects to current
  complexity.
- Do not describe remembered environmental conditions as if they are
```

```
  currently visible.
- If memory does not provide meaningful comparison, report that no substantial
  temporal change can be established.

Evidence rules:

- Use visible geometry, organization, openings, landmarks, object variety,
  visibility, and orientation cues as environmental evidence.
- Do not identify, classify, or rename spaces from uncertain contextual cues.
- Do not infer environmental quality from an uncertain place-type hypothesis.
- Do not infer personal preference, suitability, or affective response from
  environmental qualities.
- Distances are measured in tiles. Interpreted condition labels explain the
  normalized numeric values supplied by the sensing system.
- Evidence references must correspond to fields explicitly present in the
  supplied inputs.

Return ONLY one raw JSON object:
{"kaplan": {"coherence": 1-5, "complexity": 1-5,
            "legibility": 1-5, "mystery": 1-5},
 "kaplan_evidence": {
   "coherence": {"evidence": "one evidence sentence", "evidence_refs": ["input.field"]},
   "complexity": {"evidence": "one evidence sentence", "evidence_refs": ["input.field"]},
   "legibility": {"evidence": "one evidence sentence", "evidence_refs": ["input.field"]},
   "mystery": {"evidence": "one evidence sentence", "evidence_refs": ["input.field"]}},
 "environmental_qualities": {
   "summary": "one concise description of the currently perceived environmental qualities",
   "spatial_character": "short phrase",
   "orientation_support": "weak|mixed|strong",
   "salient_features": ["visible feature"]}}
```

B.2 Personal Appraiser

The Personal Appraiser interprets the perceived environment in relation to the individual's profile, including current goal, role and preference etc.

```
You are the Personal Appraiser in a spatial-experience simulation process. Interpret what the currently
     perceived environment means for this person at this moment. Use the person's specified attributes,
     current goal, current action, and recent experience to determine which perceived conditions matter and
     whether they support or constrain the current situation.

Person: !<INPUT 0>!
Role: !<INPUT 1>!
```

```
Goal: !<INPUT 2>!
Why appraisal was requested: !<INPUT 3>!
Goal-independent environmental evaluation V_t: !<INPUT 4>!
Current action and navigation state: !<INPUT 5>!
Recent interaction history: !<INPUT 6>!

Appraisal rules:
- Treat V_t as the complete environmental evidence for this appraisal. Do not request, reconstruct, or
    independently reason from underlying perceptual inputs, and do not re-evaluate the environment or
    recompute scores.
- Identify only conditions relevant to the current goal, explicitly stated profile attributes, role-related
    requirements, or ongoing action.
- Interpret relevant conditions as supporting, constraining, or having little effect on the person's current
    situation.
- The same environmental condition may have different meanings for different people only when the supplied
    profile or context supports that difference.
- goal_alignment is: helps when the situation meaningfully supports the current goal; hinders when it creates
    a meaningful obstacle or mismatch; neutral otherwise.
- appraisal_valence should follow from this person-environment relationship, not directly from environmental-
    quality scores.
- Use only explicitly supplied preferences, capabilities, needs, and prior knowledge. Do not infer traits from
     identity or role labels.
- Recent interaction history may provide temporal context from previous appraisals or actions, but it should
    not be used to reconstruct or override the current environmental evaluation \(V_t\).

Action-control rules:
- maintain_current_action when the person is still progressing along a valid route and the new evidence does
    not invalidate the current intention.
- request_new_decision when there is no executable action, the current action is finished or blocked, the
    current candidate has been inspected and a new destination is needed, or strong new evidence makes the
    current choice inappropriate.
- Ordinary visual change, mild discomfort, novelty, low confidence, or a periodic checkpoint alone is not a
    reason to abandon a valid route.
- Do not request a new decision merely because another route might also be interesting.
- The backend, not this prompt, confirms goal completion and reachability.

Use goal_progress only as an evidence-grounded description:
- en_route: still moving toward the current destination;
- at_candidate: experiencing a possible destination awaiting validation;
- target_confirmed: direct sensed evidence clearly matches the goal;
- candidate_unsuitable: direct evidence makes the inspected candidate unsuitable;
- uncertain: evidence is insufficient to classify progress.
```

```
Return ONLY one raw JSON object:
{"spatial_evaluation": "concise first-person account grounded in V_t",
 "appraisal": {
   "affect": "positive|neutral|negative",
   "goal_alignment": "helps|neutral|hinders",
   "inclination": "continue|pause|accept|reject",
   "note": "one concise first-person sentence about the present experience"},
 "goal_progress": {
   "status": "en_route|at_candidate|target_confirmed|candidate_unsuitable|uncertain",
   "reason": "one short evidence-based explanation"},
 "action_control": {
   "recommendation": "maintain_current_action|request_new_decision",
   "reason": "one short explanation tied to current action state and evidence"}}
```

B.3 Final Experience Generation

In the final generation stage, the appraisal and experiential traces are aggregated and accumulated throughout the interaction into a summary of the individual's overall spatial experience.

```
Aggregate the experiential and appraisal traces accumulated throughout the interaction into a coherent account
of the person's overall spatial experience. Do not form new environmental evaluations or contextual appraisals.


Person: !<INPUT 0>!
Identity, role, and explicitly specified attributes: !<INPUT 1>!
Goal and recorded final outcome: !<INPUT 2>!
Route and interaction statistics for this run: !<INPUT 3>!
Contextual appraisal and experiential-state trajectory, in temporal order: !<INPUT 4>!
Accumulated experience and spatial memory summaries: !<INPUT 5>!

The trajectory is the primary evidence. It may contain contextual interpretations, goal alignment, affect or
experiential state, relevant opportunities or constraints, action tendencies, and their changes over time.
Route statistics describe behavior but do not establish valence by themselves. Persona identity provides
context only and is not evidence of a preference or experience.

Grounding rules:
- Synthesize only information supported by the supplied appraisal, experience, route, and memory traces.
- Do not generate new appraisals, preferences, emotions, capabilities, or environmental conditions.
- Do not infer preferences or experiences from persona identity or role unless they are explicitly represented
      in the supplied inputs.
- Do not infer preferred or avoided areas from environmental-quality scores alone.
- Goal success does not imply a positive experience, and goal failure does not necessarily imply a negative
     experience. It need to be assessed based on the overall experience.
```

```
- Current and accumulated experiential evidence should determine the final report.
- If there is insufficient evidence for preferred_areas or avoid_areas, return [].
- Wayfinding must summarize only this run, never people of this role in general.
- The narrative may connect temporally ordered events already present in the trace, but must not invent causal
      explanations..

Return ONLY one raw JSON object, with no markdown or commentary, in exactly
this shape:
{"goal_outcome": "one sentence describing whether and how the goal was achieved",
 "experience_summary": {
   "overall_valence": "positive|mixed|neutral|negative",
   "dominant_experiences": ["experience explicitly supported by the trace"],
   "main_contributors": ["environmental or individual-context factor explicitly supported by the trace"]},
 "preferred_areas": ["area with explicit positive experiential evidence"],
 "avoid_areas": ["area with explicit negative, unsuitable, or confusing experiential evidence"],
 "wayfinding": "one sentence summarizing navigation experience during this run",
 "narrative": "3-5 first-person sentences synthesizing how the experience developed over the whole journey"}
```

B.4 Experience-Metric Prompts

At the end of each simulation episode, the model rated the applicable experience metrics on a 7-point scale. The exact question prompt and scale anchors are provided below.

Common Instruction

Complete the supplied structured experience questionnaire as a self-report.
This questionnaire is independent of the previous Final Experience synthesis and must use only the evidence supplied below. Return one integer score from 1 to 7 for every question and apply its supplied anchors exactly.

Goal Support

To what extent did the encountered environment support the individual's current goal across the episode?
Anchors: 1 = strongly hindered the goal; 7 = strongly supported the goal.

Overall Experience

How positive or negative was the individual's overall spatial experience across the episode?
Anchors: 1 = very negative; 7 = very positive.

Navigation Confidence

How confident was the individual about understanding where to go during the task?
Anchors: 1 = very uncertain about where to go; 7 = very confident about where to go.

Route Clarity

How clearly did the building's layout itself show the way to the destination during this run?
Anchors: 1 = the way was never apparent and the destination was found by trial and error;

```
7 = the way was obvious from the corridors, sightlines, and door signs.

Navigation Control
To what extent did the individual feel able to understand and manage the navigation process?
Anchors: 1 = very little sense of control; 7 = very strong sense of control.

Task Control
To what extent did the individual feel able to accomplish the current activity under the encountered
spatial conditions?
Anchors: 1 = very little sense of control; 7 = very strong sense of control.

Task Suitability
How suitable was the encountered environment for the individual's current intended activity?
Anchors: 1 = very unsuitable; 7 = very suitable.

Visual Comfort
How visually comfortable were the encountered environmental conditions for this individual?
Anchors: 1 = very visually uncomfortable; 7 = very visually comfortable.

Preference Match
How well did the encountered environmental conditions match the individual's explicitly specified
environmental preference?
Anchors: 1 = strongly mismatched; 7 = strongly matched.

Ease of Movement
How easy was it for the individual to move through the environment while carrying out the current task?
Anchors: 1 = very difficult; 7 = very easy.

Task Stress
How stressful were the encountered spatial and social conditions while carrying out the current task?
Anchors: 1 = not stressful; 7 = very stressful.
```

C Full Statistical Results

This appendix reports the complete quantitative results for the five comparative scenarios. Table 4 reports the primary experience and task-process measures, while Table 5 reports the Kaplan-based outputs of the Environment Evaluator as intermediate measures. The statistical procedures used for these comparisons are described in Section 6.2.

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Table 4. Full results for the experience and task-process measures across the five comparative scenarios.

Scenario	Type	Metric	Group A M (SD)	Group B M (SD)	MD [95% CI]	Welch p_{adj}	Perm. p_{adj}
Layout	Experience	Goal Support	6.500 (0.527)	4.200 (1.989)	2.300 [1.100, 3.500]	0.005	0.007
		Overall Experience	6.000 (0.667)	3.800 (1.549)	2.200 [1.200, 3.200]	0.004	0.007
		Navigation Confidence	5.500 (0.707)	3.800 (1.229)	1.700 [0.900, 2.500]	0.004	0.007
		Route Clarity	5.200 (0.632)	3.800 (0.919)	1.400 [0.800, 2.100]	0.004	0.007
	Process	Steps to Completion	99.500 (20.823)	223.000 (0.000)	−123.500 [−135.000, −112.000]	< 0.001	< 0.001
		Candidates Rejected	2.500 (0.527)	11.000 (0.000)	−8.500 [−8.800, −8.200]	< 0.001	< 0.001
		Path Efficiency	0.650 (0.139)	0.660 (0.000)	−0.010 [−0.088, 0.069]	0.829	0.763
Memory	Experience	Goal Support	6.700 (0.483)	2.000 (0.000)	4.700 [4.400, 5.000]	< 0.001	< 0.001
		Overall Experience	6.000 (0.667)	2.000 (0.000)	4.000 [3.600, 4.400]	< 0.001	< 0.001
		Navigation Confidence	6.400 (0.516)	2.100 (0.316)	4.300 [3.900, 4.600]	< 0.001	< 0.001
		Navigation Control	6.200 (0.422)	2.200 (0.422)	4.000 [3.700, 4.300]	< 0.001	< 0.001
	Process	Steps to Completion	94.000 (0.000)	177.000 (0.000)	−83.000 [−83.000, −83.000]	N/A	< 0.001
		Candidates Rejected	0.000 (0.000)	1.400 (2.366)	−1.400 [−3.000, −0.400]	0.094	0.003
		Path Efficiency	1.000 (0.000)	0.550 (0.000)	0.450 [0.450, 0.450]	N/A	< 0.001
Role	Experience	Goal Support	6.600 (0.516)	3.700 (1.636)	2.900 [1.900, 3.900]	< 0.001	< 0.001
		Overall Experience	6.400 (0.516)	3.300 (1.418)	3.100 [2.200, 4.000]	< 0.001	< 0.001
		Task Control	6.300 (0.483)	4.700 (1.252)	1.600 [0.800, 2.400]	0.003	0.001
		Task Suitability	6.600 (0.516)	3.700 (1.418)	2.900 [2.000, 3.800]	< 0.001	< 0.001
	Process	Steps to Completion	123.000 (0.000)	60.200 (5.903)	62.800 [60.000, 67.000]	< 0.001	< 0.001
		Candidates Rejected	0.000 (0.000)	3.400 (1.430)	−3.400 [−4.200, −2.500]	< 0.001	< 0.001
		Path Efficiency	1.000 (0.000)	0.815 (0.097)	0.185 [0.115, 0.231]	< 0.001	0.001
Weather	Experience	Goal Support	7.000 (0.000)	6.600 (0.516)	0.400 [0.100, 0.700]	0.090	0.193
		Overall Experience	6.700 (0.483)	6.300 (0.675)	0.400 [−0.100, 0.900]	0.147	0.272
		Visual Comfort	6.100 (0.316)	5.500 (0.707)	0.600 [0.100, 1.000]	0.090	0.193
		Preference Match	6.900 (0.316)	6.000 (0.816)	0.900 [0.400, 1.400]	0.029	0.056
	Process	Steps to Completion	38.800 (4.367)	50.200 (7.955)	−11.400 [−16.600, −6.200]	0.004	0.004
		Candidates Rejected	0.000 (0.000)	1.000 (1.054)	−1.000 [−1.600, −0.400]	0.030	0.072
		Path Efficiency	1.000 (0.000)	0.890 (0.124)	0.110 [0.035, 0.185]	0.030	0.072
Occupancy	Experience	Goal Support	6.400 (0.516)	5.400 (0.966)	1.000 [0.400, 1.700]	0.012	0.009
		Overall Experience	5.900 (0.316)	3.900 (0.994)	2.000 [1.400, 2.600]	< 0.001	< 0.001
		Ease of Movement	6.000 (0.000)	2.900 (0.994)	3.100 [2.500, 3.600]	< 0.001	< 0.001
		Task Stress	2.000 (0.000)	4.500 (0.850)	−2.500 [−3.000, −2.000]	< 0.001	< 0.001
	Process	Steps to Completion	119.000 (0.000)	119.000 (0.000)	0.000 [0.000, 0.000]	N/A	N/A
		Candidates Rejected	0.000 (0.000)	0.000 (0.000)	0.000 [0.000, 0.000]	N/A	N/A
		Path Efficiency	1.000 (0.000)	1.000 (0.000)	0.000 [0.000, 0.000]	N/A	N/A

Group definitions. Layout: A = Direct, B = Indirect; Memory: A = Familiar, B = Unfamiliar; Role: A = Professor, B = Student; Weather: A = Cloudy, B = Sunny; Occupancy: A = Low Occupancy, B = High Occupancy. MD = A–B; CI = 95% bootstrap confidence interval. Welch and permutation p-values are Holm-adjusted separately within the experience and task-process measure families for each scenario. All statistics are rounded to three decimal places.

Table 5. Intermediate Kaplan evaluation results across the five comparative scenarios.

Scenario	Metric	Group A M (SD)	Group B M (SD)	MD [95% CI]	Welch p_{adj}	Perm. p_{adj}
Layout	Coherence	4.029 (0.113)	4.406 (0.216)	−0.378 [−0.519, −0.231]	0.001	< 0.001
	Complexity	3.634 (0.217)	3.418 (0.143)	0.217 [0.074, 0.378]	0.054	0.035
	Legibility	4.154 (0.180)	4.400 (0.329)	−0.246 [−0.462, −0.020]	0.115	0.114
	Mystery	4.603 (0.235)	4.541 (0.274)	0.062 [−0.148, 0.271]	0.597	0.591
Memory	Coherence	4.720 (0.253)	3.729 (0.267)	0.991 [0.772, 1.203]	< 0.001	< 0.001
	Complexity	3.820 (0.257)	4.036 (0.148)	−0.216 [−0.396, −0.041]	0.037	0.036
	Legibility	4.780 (0.199)	3.157 (0.240)	1.623 [1.438, 1.800]	< 0.001	< 0.001
	Mystery	2.400 (0.211)	4.993 (0.022)	−2.593 [−2.726, −2.473]	< 0.001	< 0.001
Role	Coherence	4.680 (0.253)	4.200 (0.230)	0.480 [0.270, 0.670]	< 0.001	0.001
	Complexity	3.920 (0.169)	3.542 (0.252)	0.378 [0.208, 0.563]	0.001	< 0.001
	Legibility	4.780 (0.220)	4.125 (0.212)	0.655 [0.470, 0.830]	< 0.001	< 0.001
	Mystery	2.540 (0.212)	3.100 (0.280)	−0.560 [−0.772, −0.355]	< 0.001	< 0.001
Weather	Coherence	4.350 (0.337)	4.350 (0.211)	0.000 [−0.225, 0.250]	1.000	1.000
	Complexity	3.650 (0.242)	3.325 (0.206)	0.325 [0.150, 0.525]	0.019	0.042
	Legibility	4.500 (0.333)	4.525 (0.362)	−0.025 [−0.325, 0.250]	1.000	1.000
	Mystery	3.650 (0.530)	3.800 (0.258)	−0.150 [−0.500, 0.200]	1.000	1.000
Occupancy	Coherence	4.725 (0.184)	4.750 (0.204)	−0.025 [−0.175, 0.125]	1.000	1.000
	Complexity	3.550 (0.158)	3.675 (0.169)	−0.125 [−0.250, 0.000]	0.419	0.792
	Legibility	4.675 (0.206)	4.625 (0.177)	0.050 [−0.125, 0.200]	1.000	1.000
	Mystery	2.250 (0.204)	2.250 (0.204)	0.000 [−0.175, 0.175]	1.000	1.000

Group definitions. Layout: A = Direct, B = Indirect; Memory: A = Familiar, B = Unfamiliar; Role: A = Professor, B = Student; Weather: A = Cloudy, B = Sunny; Occupancy: A = Low Occupancy, B = High Occupancy. MD = A–B; CI = 95% bootstrap confidence interval. Welch and permutation p-values are Holm-adjusted separately within the experience and task-process measure families for each scenario. All statistics are rounded to three decimal places.